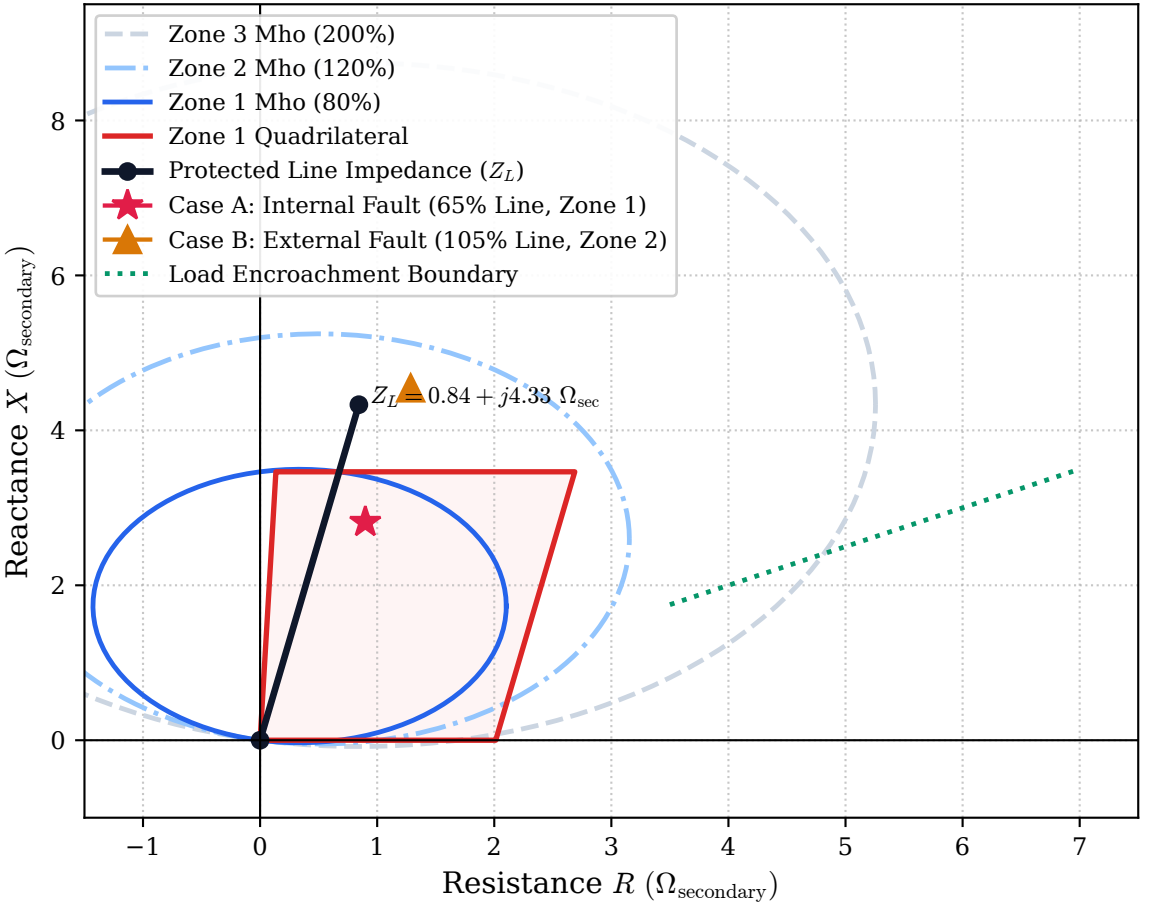
ANSI 21 Distance Relay Multi-Zone Characteristics ($R-X$ Plane)
Zone 3 Mho (200%)
Zone 2 Mho (120%)
Zone 1 Mho (80%)
Zone 1 Quadrilateral
Protected Line Impedance (Z_L)
Case A: Internal Fault (65% Line, Zone 1)
Case B: External Fault (105% Line, Zone 2)
Load Encroachment Boundary
$Z_L = 0.84 + j4.33\ \Omega_{\text{sec}}$
Reactance X ($\Omega_{\text{secondary}}$)
Resistance R ($\Omega_{\text{secondary}}$)
8
6
4
2
0
−1
0
1
2
3
4
5
6
7